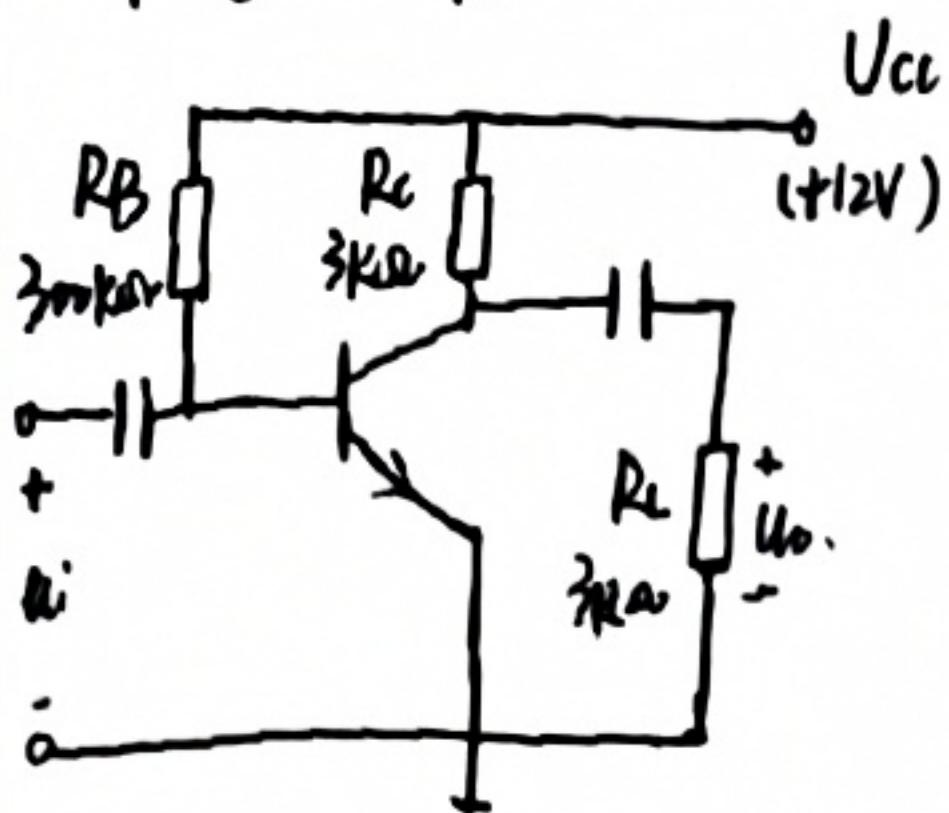
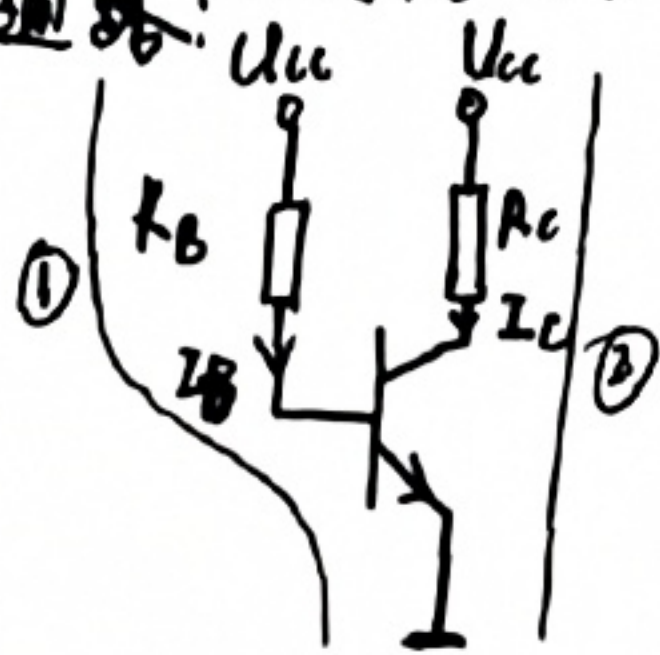


1.  $\beta=50$ . 画直流通路和计算  $r_{be}$ , 静态工作点.



直流通路: (电容视为开路).



$U_{CC}=12V, R_B=300k\Omega, R_C=3k\Omega, \beta=50, U_{BEQ}=0.7V$

① 回路:  $U_{CC} = R_B \cdot I_{BQ} + U_{BEQ}$

$$I_{BQ} = \frac{U_{CC} - U_{BEQ}}{R_B} = 38 \mu A$$

$$I_{CQ} = \beta I_{BQ} = 1.9 \text{ mA}$$

② 回路:  $U_{CC} = I_{CQ} \cdot R_C + U_{CEQ}$

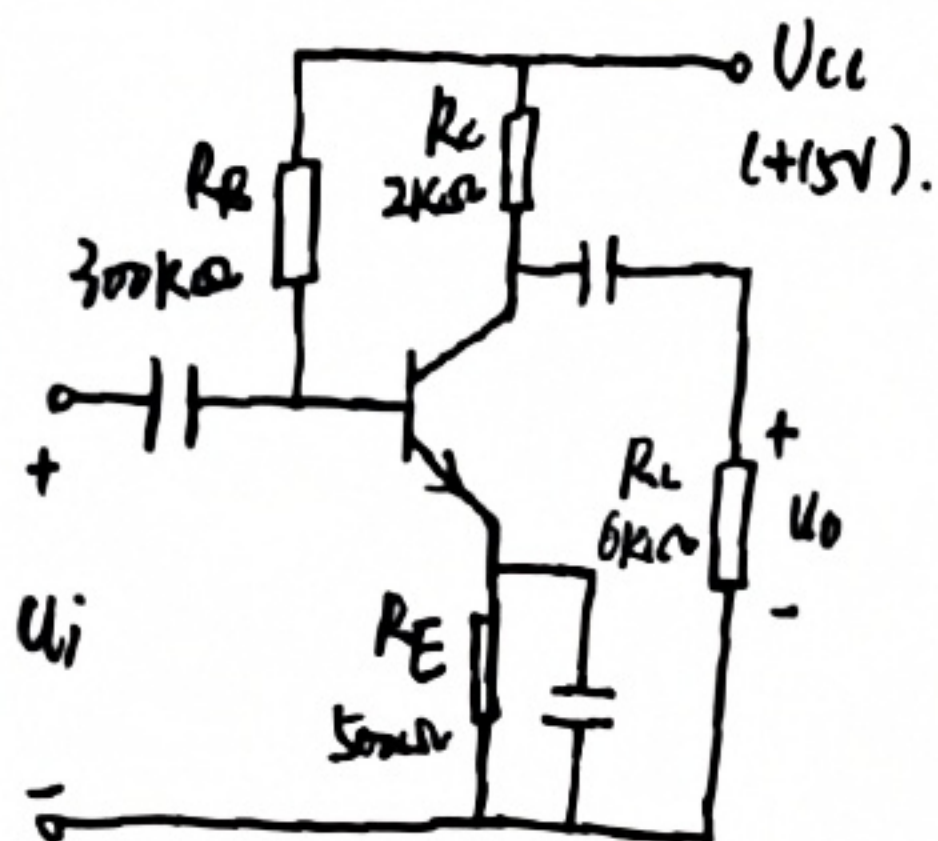
$$U_{CEQ} = U_{CC} - I_{CQ} \cdot R_C = 6.3V$$

$$I_{EQ} = (1 + \beta) I_{BQ}, r_{be} = r_{bb'} + (1 + \beta) \frac{26 \text{ mV}}{I_{EQ}}$$

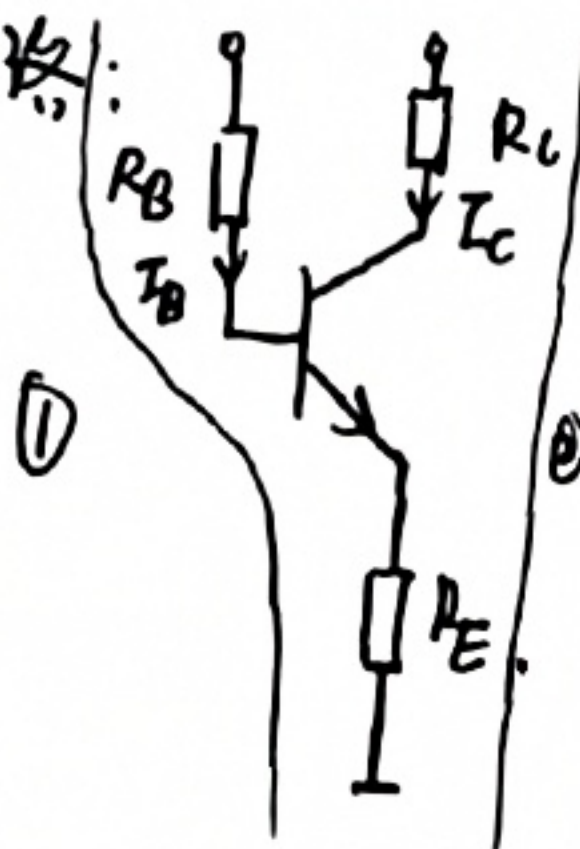
$$\text{即 } r_{be} = 384 \Omega$$

(注:  $r_{bb'}$  题目中没有任何提示, 默认为  $300 \Omega$ )

2.  $\beta=100$ , 画直流通路, 静态工作点.



直流通路:



① 回路:  $U_{CC} = R_B I_{BQ} + U_{BEQ} + R_E \cdot I_{EQ}$

② 回路:  $U_{CC} = I_{CQ} R_C + U_{CEQ} + I_{EQ} \cdot R_E$

$$I_{CQ} = \beta I_{BQ}$$

$$I_{EQ} = (1 + \beta) I_{BQ}$$

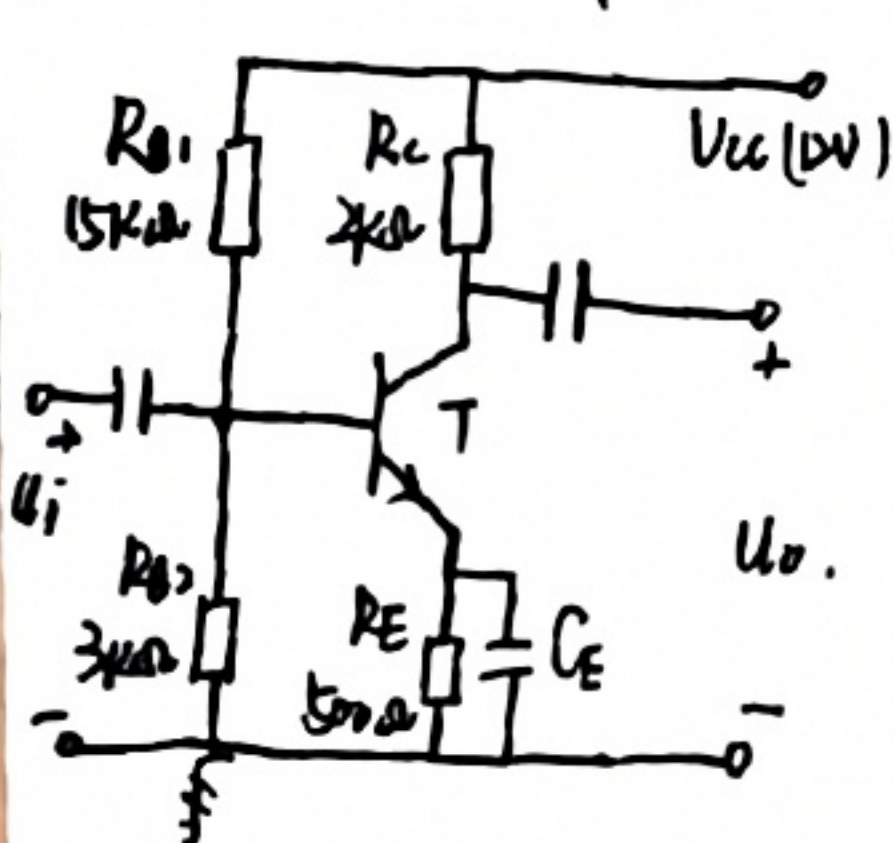
联立 ① ② ③ ④ 可得  $I_{BQ} \approx 43 \mu A$

$$I_{CQ} = 4.3 \text{ mA}$$

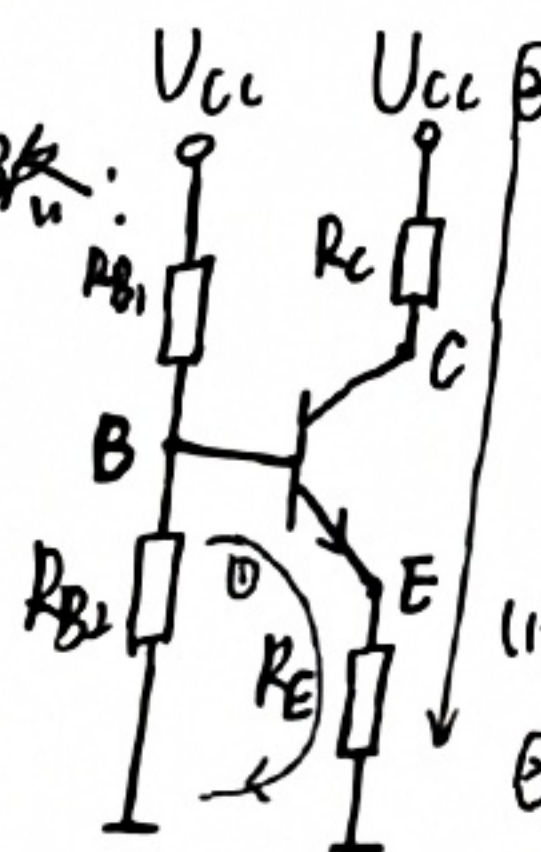
$$I_{EQ} =$$

$$U_{CEQ} = 4.3V$$

3.  $\beta=40, U_{BE}=0.7V$ .



直流通路:



$$U_{BQ} = \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot U_{CC} = \frac{3}{15 + 2} \times 12V = 2V$$

①:  $I_{EQ} \cdot R_E = U_B - U_{BEQ} \Rightarrow I_{EQ} = \frac{U_B - U_{BEQ}}{R_E} = 2.6 \text{ mA}$

$I_{CQ} = 2.6 \text{ mA}$  (近似等于  $I_{EQ}$ )

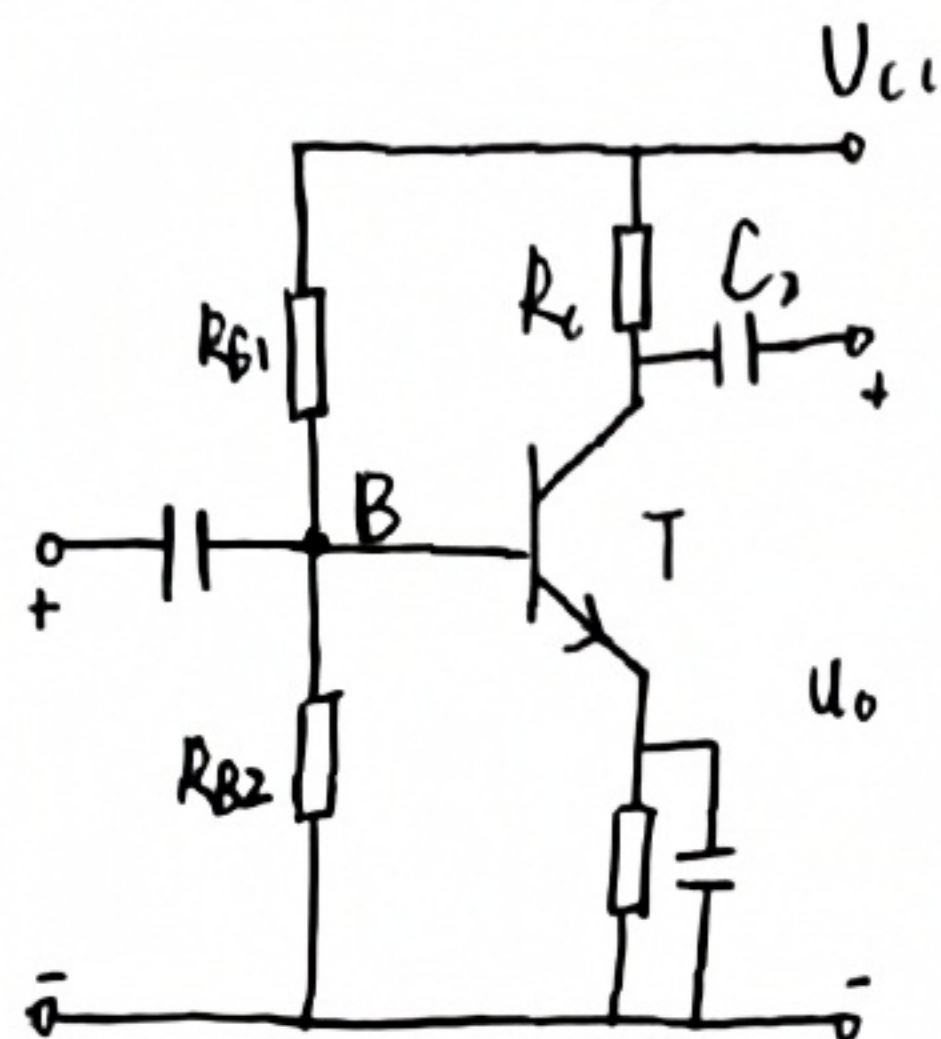
(1)  $I_{BQ} = I_{EQ} \Rightarrow I_{BQ} = \frac{I_E}{1 + \beta} = 63 \mu A$

②:  $U_{CC} = U_{CEQ} + I_C \cdot (R_C + R_E) \Rightarrow U_{CEQ} = U_{CC} - I_C \cdot (R_C + R_E)$   
 $= 12 - 2.6 \times (2 + 0.5)$   
 $= 5.5V$

4. 同第2题一样.



5.  $R_{B1}$  或  $R_{B2}$  为热敏电阻, 当  $T$  上升时,  $I_{CQ}$ ,  $U_{EQ}$ ,  $U_{CEQ}$  的变化情况。



$U_{CC} = I_{B1} \cdot R_{B1} + I_{B2} \cdot R_{B2}$  ① 为  $R_{B1}$  为热敏电阻:

$U_{CC} = I_B \cdot R_{B1} + U_{BE} + I_E R_E$   $T \uparrow \rightarrow R_{B1} \downarrow \xrightarrow{R_{B1} \downarrow \rightarrow U_B \uparrow} U_{BE} \uparrow$

$U_{CC} = I_C \cdot R_C + U_{CE} + I_E \cdot R_E \rightarrow I_B \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow \rightarrow U_{EQ} \uparrow$

$I_E = (1 + \beta) I_B$

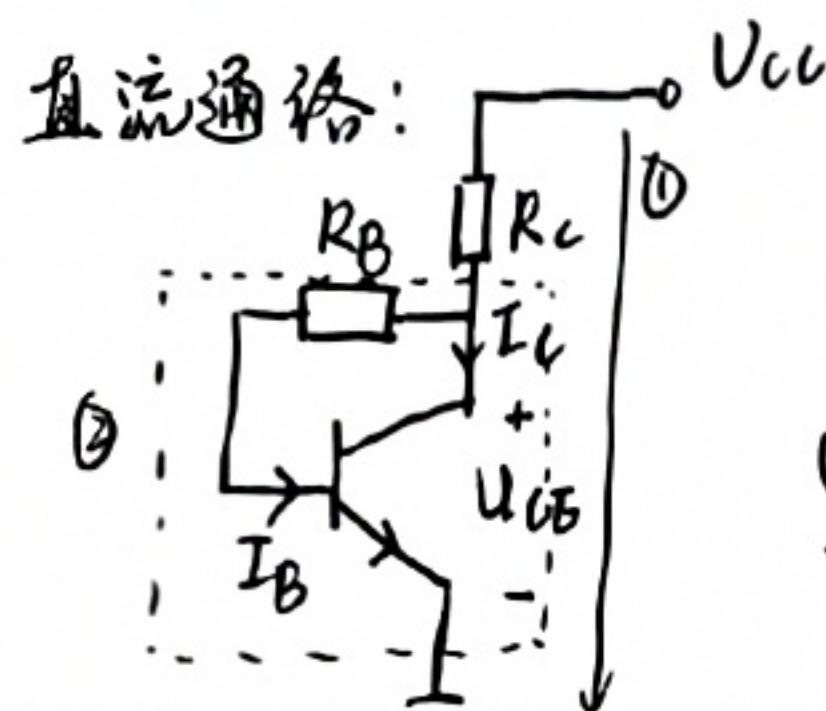
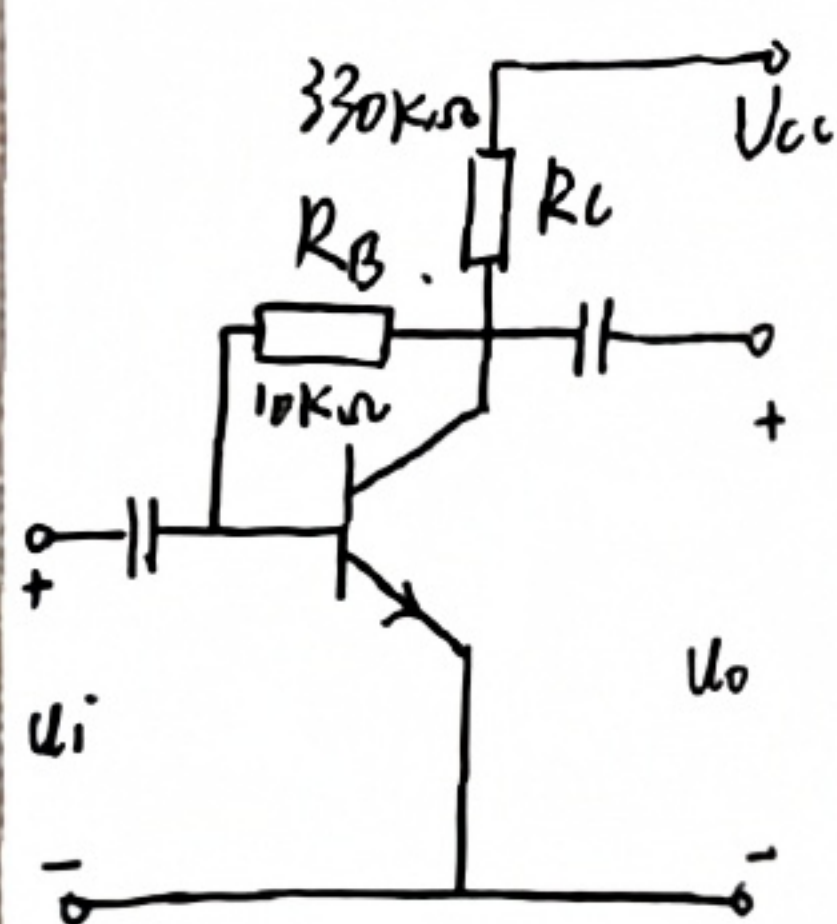
$I_C = \beta I_B$

② 为  $R_{B2}$  为热敏电阻时:

$T \uparrow \rightarrow R_{B2} \downarrow \xrightarrow{R_{B2} \downarrow \rightarrow U_B \downarrow} U_{BE} \downarrow \rightarrow I_B \downarrow$

$\rightarrow I_C \downarrow \rightarrow I_E \downarrow \rightarrow U_{EQ} \downarrow$

6.  $\beta = 50$ ,  $U_{CC} = 20V$ ,  $R_C = 10k\Omega$ ,  $R_B = 330k\Omega$ , 微变等效电路和交流通路, 负载  $R_L = 2k\Omega$ .



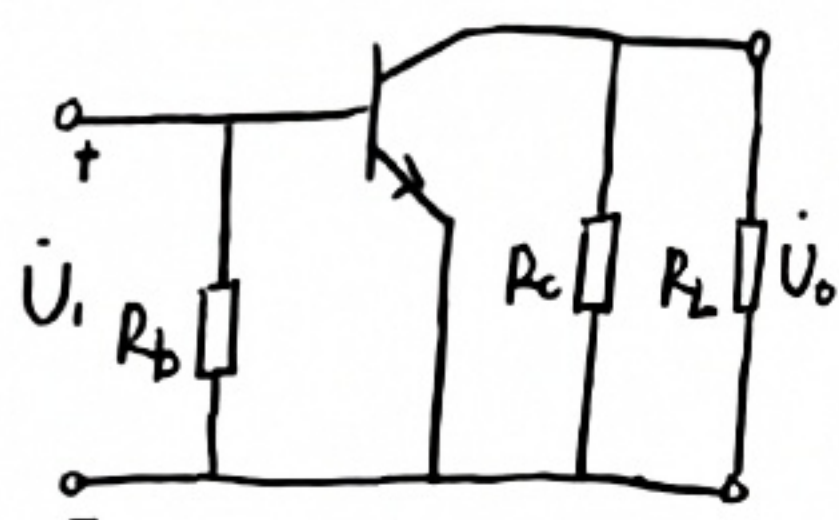
由 KVL 可得:

①  $U_{CC} = R_C \cdot (I_B + I_C) + U_{CEQ}$

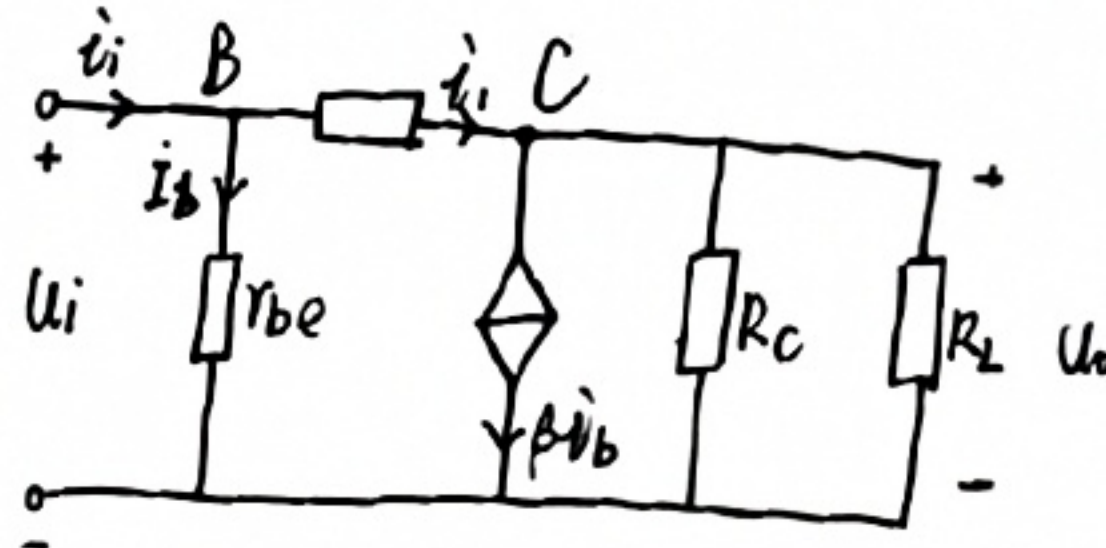
②  $U_{CE} = R_B \cdot I_B + U_{BEQ}$

可得  $I_{BQ} = 23\mu A$ ,  $I_{CQ} = 1.15mA$ ,  $U_{CEQ} = 8.3V$ .

交流通路:



微变等效电路:



$r_{be} = r_{bb'} + \beta \frac{26}{I_C} = 1.43k\Omega$

由图可得  $U_i = r_{be} \cdot I_b$  ①,  $U_{CB} = U_i - U_o$ ,  $i_1 = \frac{U_{CB}}{R_B}$  ②

$U_o = K_L' (i_1 - \beta I_b) = K_L' \cdot (\frac{U_i - U_o}{R_B} - \beta I_b)$  ③,  $K_L' = R_C \parallel R_L$  ④

联立 ①②③④ 可得  $A_u = \frac{U_o}{U_i} = \frac{K_L' (r_{be} - \beta R_B)}{r_{be} (R_L' + R_B)} \approx -\beta \frac{R_L'}{r_{be}} = -58$ .

输入电阻:

$I_i = I_b + I_1 = I_b + \beta I_b + \frac{U_o}{R_L'}$

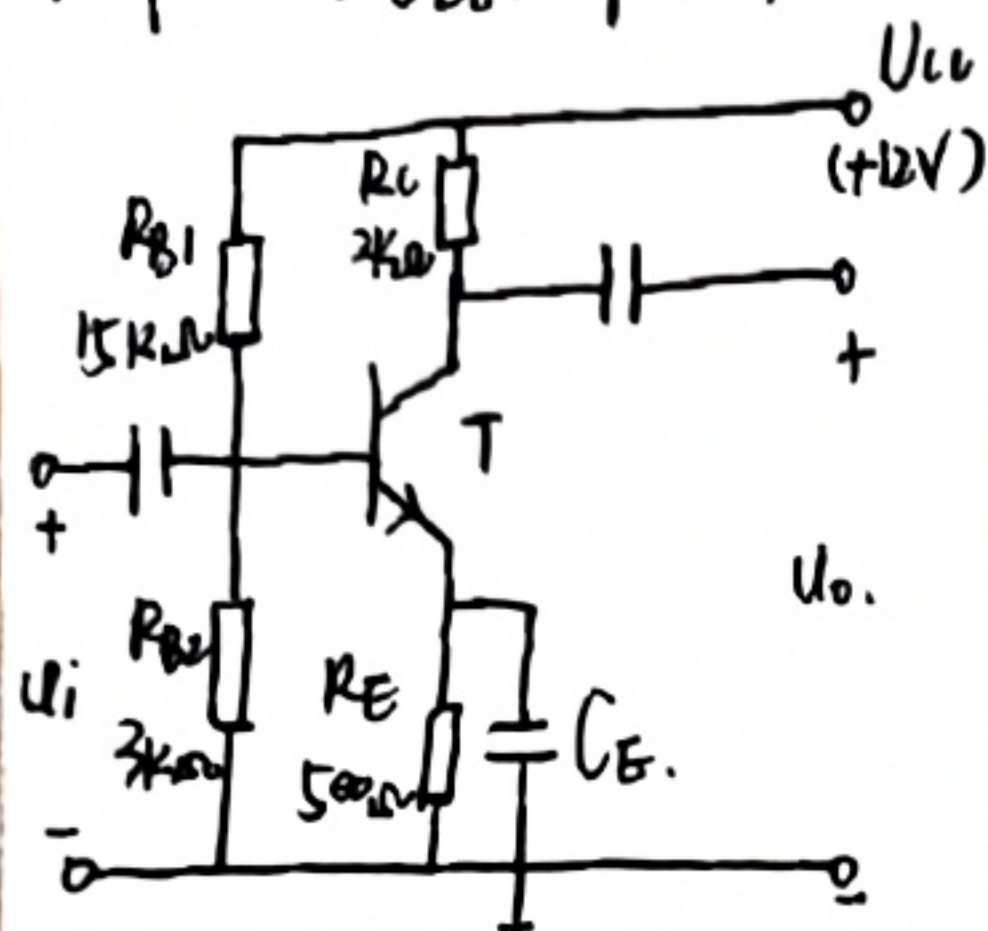
$= (1 + \beta) I_b + \frac{A_u U_i}{R_L'}$

$= (1 + \beta) \frac{U_i}{r_{be}} + \frac{(r_{be} - \beta R_B) U_i}{(R_L' + R_B) r_{be}}$

$R_i = \frac{U_i}{I_i} = \frac{R_B r_{be}}{R_B + (1 + \beta) R_L'} = 1.14k\Omega$



7.  $\beta=40, U_{BE}=0.7V$ , 微变等效电路和交流通路,  $A_u, r_i, r_o$ .



由第(3)题求得. 交流通路.

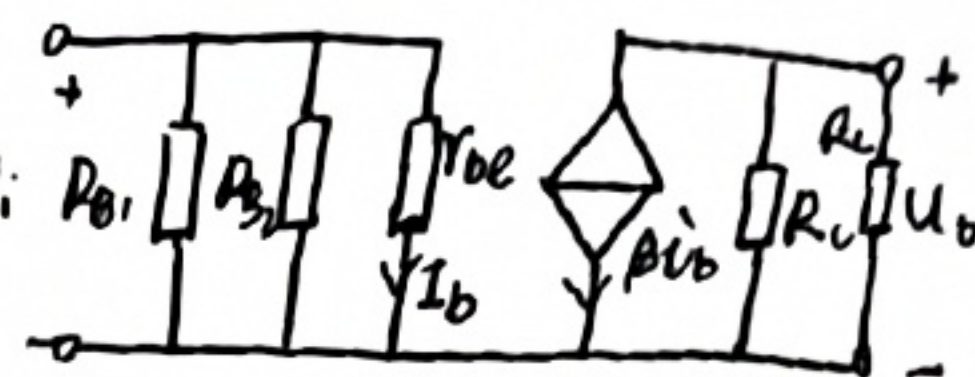
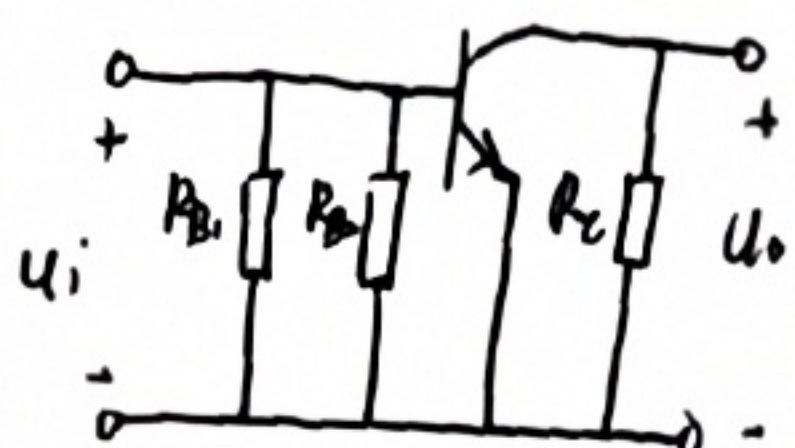
$$U_{BE}=2V$$

$$I_{EQ}=2.6mA$$

$$I_C=2.6mA$$

$$I_{BQ}=63\mu A$$

微变等效电路:



$$r_{be} = r_{bb'} + \frac{U_T}{I_{EQ}}(1+\beta) = 0.7k\Omega$$

$$A_u = \frac{U_o}{U_i} = \frac{-\beta i_b (R_C \parallel R_L)}{I_b \cdot r_{be}} = -40 \frac{215}{0.7} = -82$$

$$r_i = R_{B1} \parallel R_{B2} \parallel r_{be} = 0.55k\Omega$$

$$r_o = R_C = 2k\Omega$$

8.  $\beta=50, A_u, r_i, r_o$ .

由第1题求得:

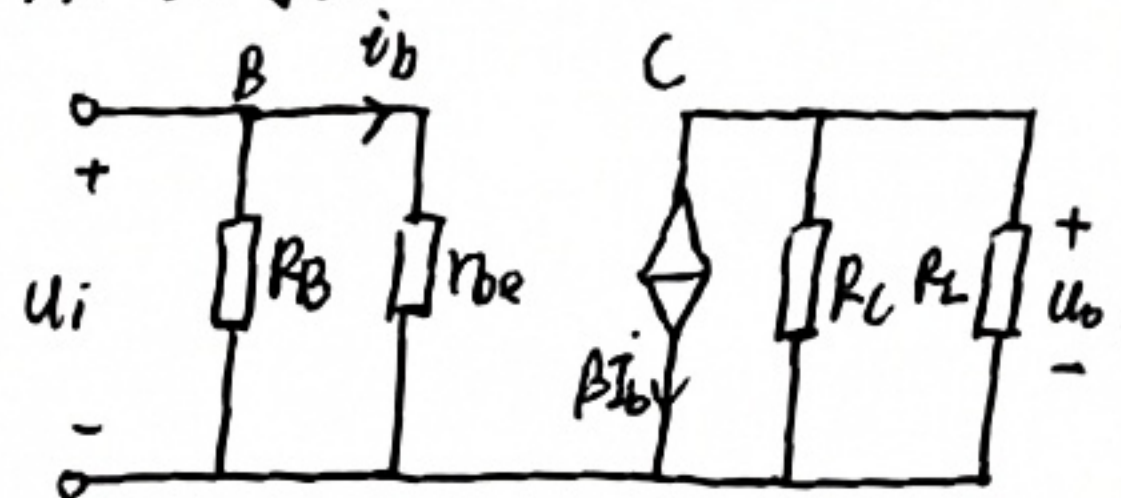
$$I_B = 38\mu A$$

$$I_{CQ} = 1.9mA$$

$$U_{CE} = 6V$$

$$r_{be} = 984\Omega$$

微变等效电路:



$$A_u = \frac{-\beta I_b (R_C \parallel R_L)}{r_{be} \cdot I_b}$$

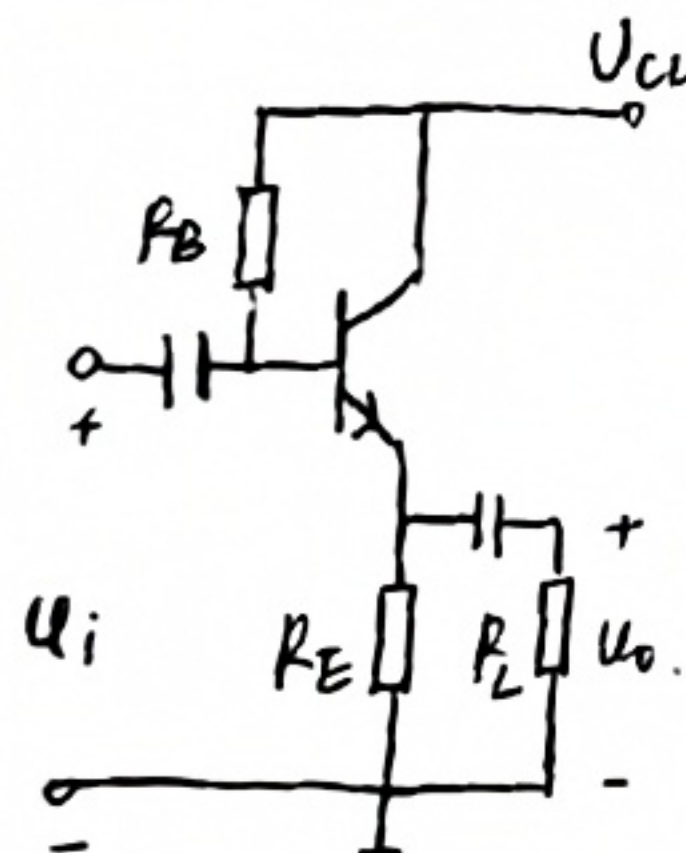
$$= -50 \frac{3113}{0.984}$$

$$= -76$$

$$r_i = R_B \parallel r_{be} = 981\Omega$$

$$r_o = R_C = 3k\Omega$$

9.  $U_{CC}=10V, R_E=5.6k\Omega, R_B=240k\Omega, \beta=40, R_S=10k\Omega, R_L=1k\Omega$ . 求  $A_u, r_i, r_o$ .



输入回路:

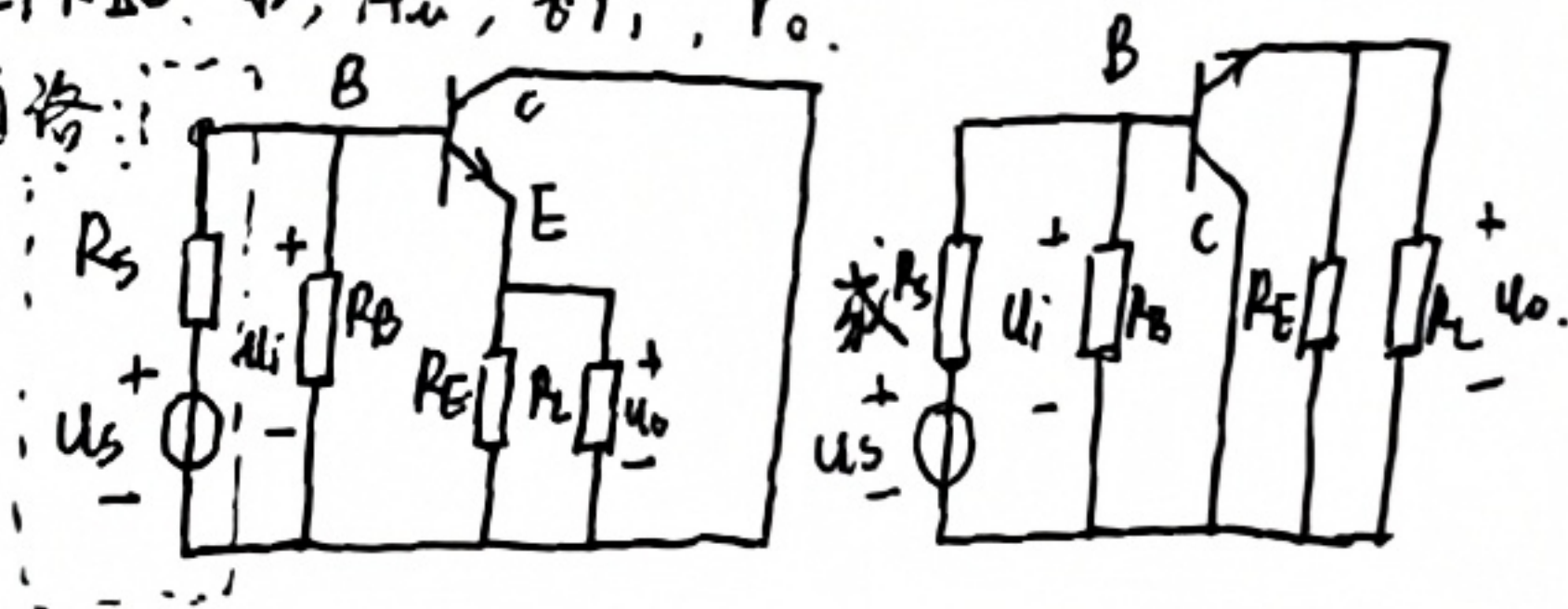
$$U_{CC} = R_B \cdot I_{BQ} + 0.7 + R_E (1+\beta) I_{BQ}$$

$$\Rightarrow I_{BQ} = 19.8\mu A$$

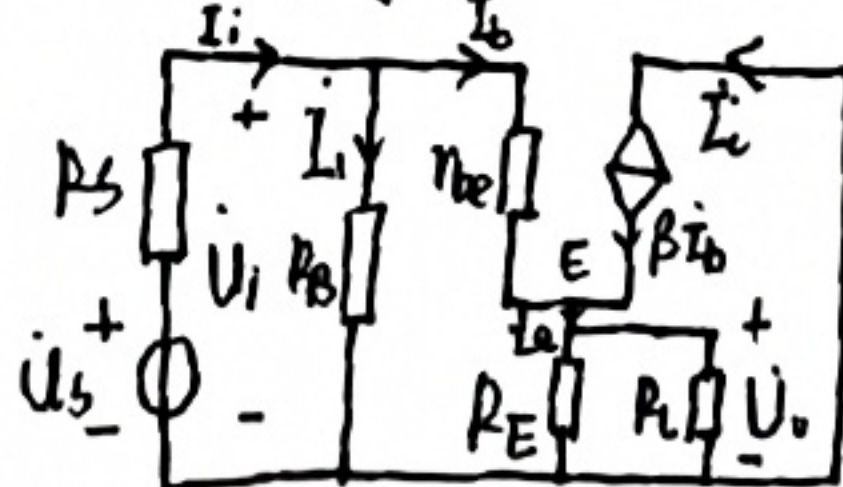
$$I_{CQ} = \beta I_{BQ} = 0.79mA$$

$$U_{CEQ} = U_{CC} - R_E \cdot I_E = 5.6V$$

交流通路:



微变等效电路:



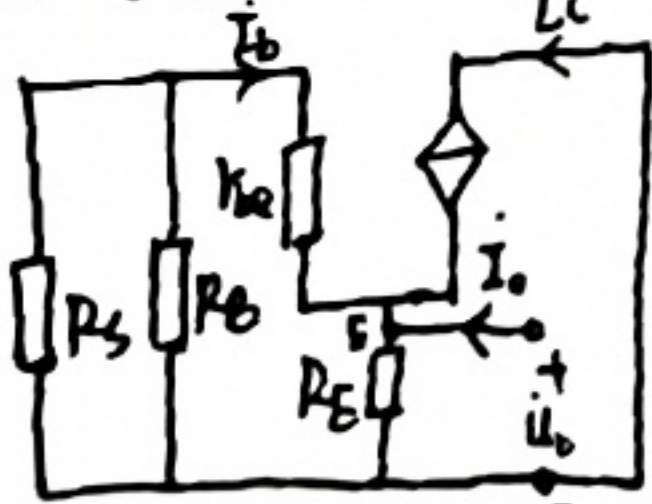
$$U_o = I_e R_L' = (1+\beta) I_b R_L'$$

$$U_i = I_b r_{be} + U_o = I_b r_{be} + (1+\beta) I_b R_L'$$

$$A_u = \frac{U_o}{U_i} = \frac{(1+\beta) R_L'}{r_{be} + (1+\beta) R_L'} \approx 1$$

$$R_L' = R_E \parallel R_L$$

计算输出电阻电路:



输入电阻:

$$I_b = \frac{U_i}{r_{be} + (1+\beta) R_L'}$$

$$I_i = I_1 + I_b = \frac{U_i}{R_B} + \frac{U_i}{r_{be} + (1+\beta) R_L'}$$

$$R_i = \frac{U_i}{I_i} \Rightarrow \frac{1}{R_i} = \frac{1}{R_B} + \frac{1}{r_{be} + (1+\beta) R_L'}$$

$$\Rightarrow R_i = R_B \parallel [r_{be} + (1+\beta) R_L'] = 32k\Omega$$

输出电阻: 从输出端看入, 除电源外的电阻.

$$KVL: I_b (R_S \parallel R_B) + I_b r_{be} + U_o = 0$$

$$I_b = -\frac{U_o}{r_{be} + R_S'}$$

$$R_S' = R_S \parallel R_B$$

$$KCL: I_{RE} = I_b + \beta I_b + I_o \quad (I_{RE} \text{ 为 } RE \text{ 支路电流})$$

$$I_{RE} \cdot R_E = U_o$$

$$\beta I_b = I_{RE} - (1+\beta) I_b$$

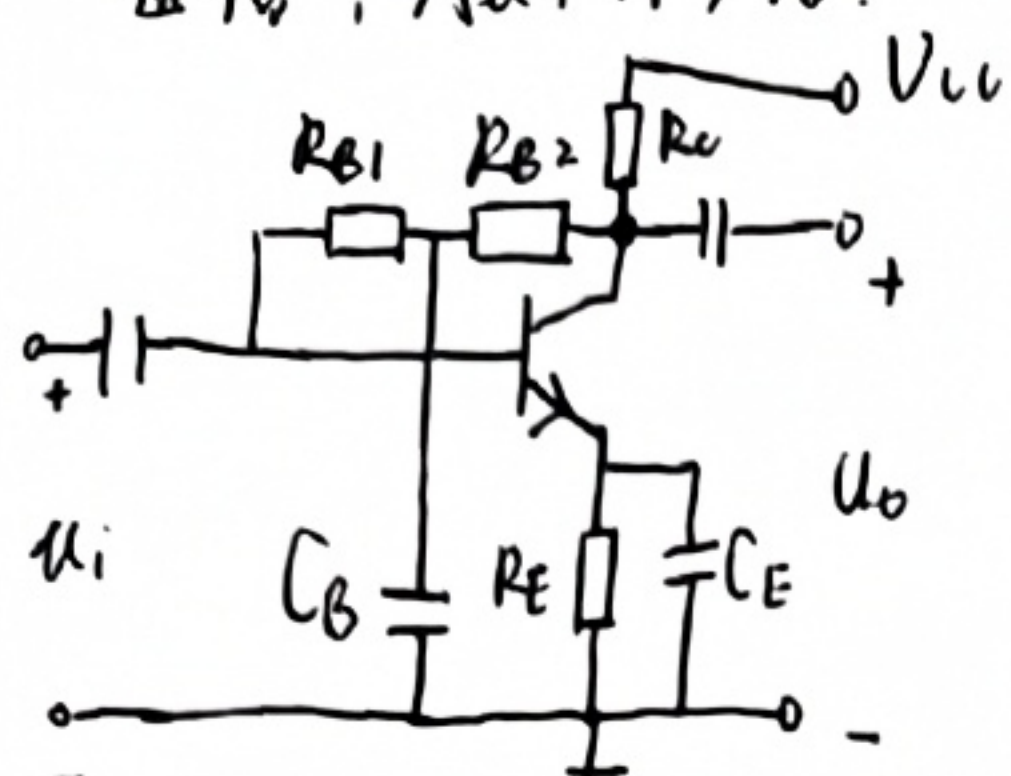
$$= \frac{U_o}{R_E} + \frac{U_o \cdot (1+\beta)}{r_{be} + R_S'}$$

$$\frac{1}{R_o} = \frac{I_o}{U_o} = \frac{1}{R_E} + \frac{1}{\frac{r_{be} + R_S'}{1+\beta}}$$

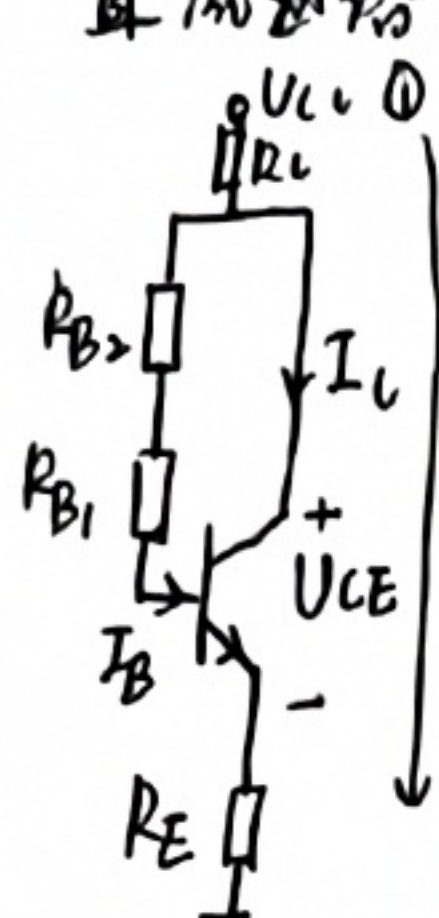
$$R_o = R_E \parallel \frac{r_{be} + R_S'}{1+\beta} = 0.27k\Omega$$



10.  $R_{B1} = 75k\Omega$ ,  $R_{B2} = 5k\Omega$ ,  $R_C = 4k\Omega$ ,  $R_E = 1k\Omega$ ,  $\beta = 50$ ,  $U_{BE} = 0.7V$ ,  $U_{CC} = 12V$ . 分压式等效电路, 交流通路,  $A_u$ ,  $r_i$ ,  $r_o$ .



直流通路:



由KVL可得:

$$U_{CC} = R_C(I_{BQ} + I_{CQ}) + U_{CEQ} + R_E \cdot I_{EQ}$$

$$U_{CEQ} = (R_{B1} + R_{B2})I_{BQ} + U_{BEQ}$$

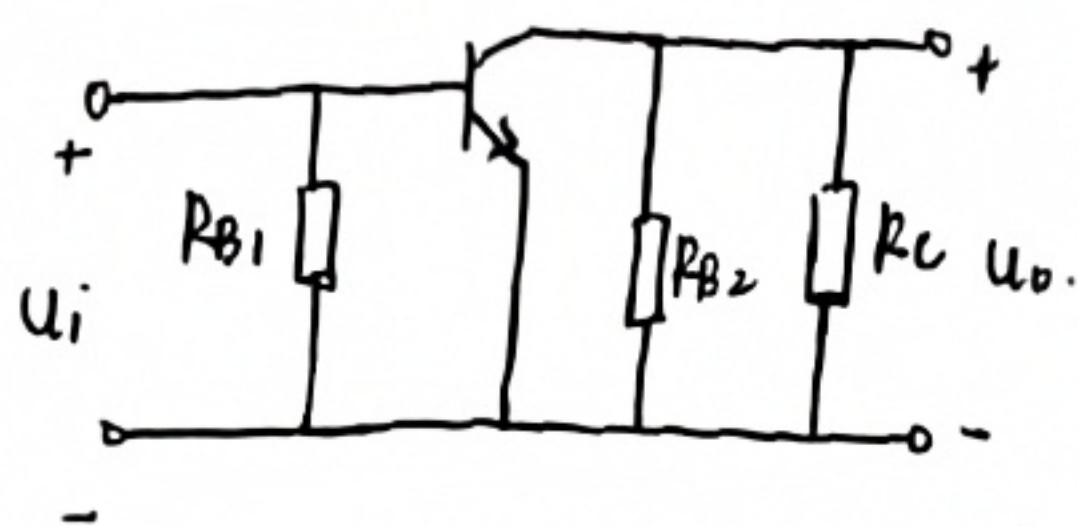
$$I_{CQ} = \beta I_{BQ} \quad U_{BEQ} = 0.7V$$

$$\text{可得 } I_{BQ} = 34\mu A$$

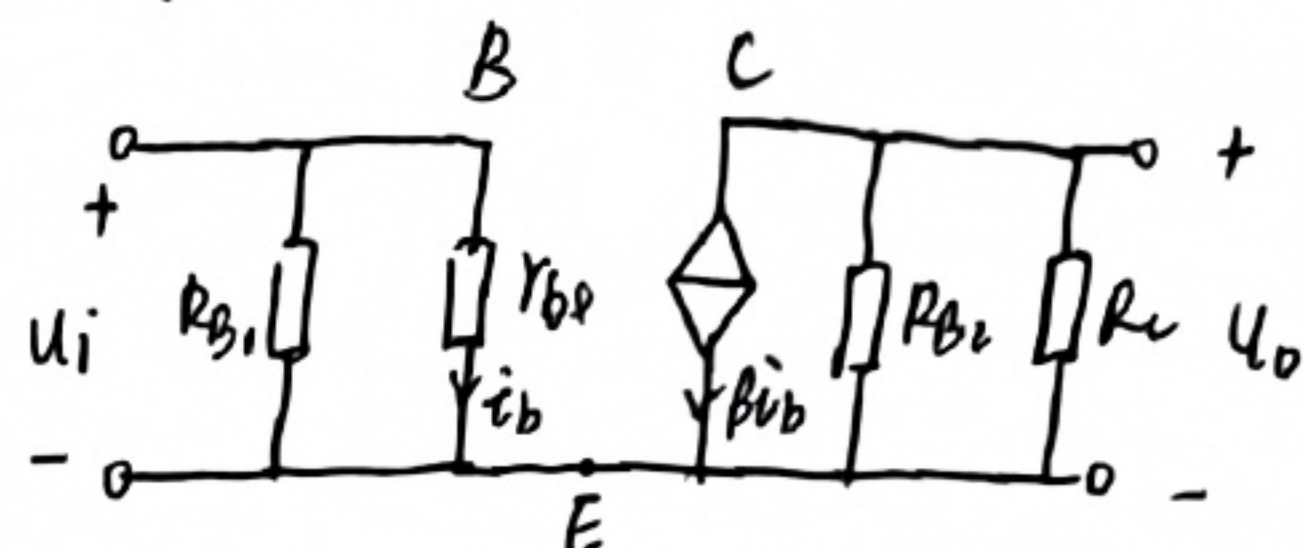
$$I_{CQ} = 1.7mA$$

$$U_{CEQ} = 3.5V$$

交流通路:



微变等效电路:

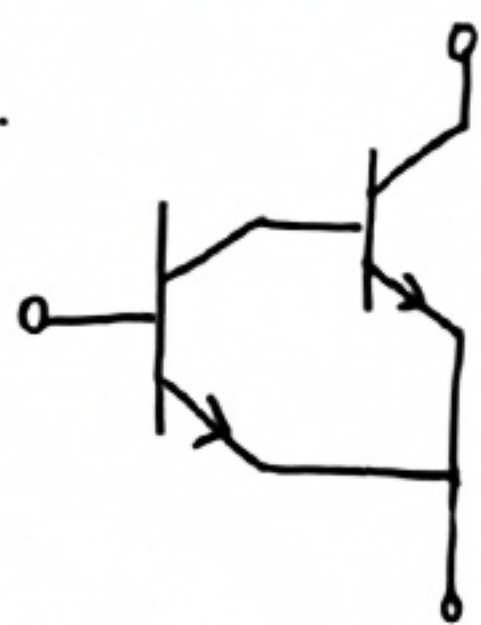


$$A_u = \frac{u_o}{u_i} = \frac{-\beta i_b (R_{B2} \parallel R_C)}{r_{be} \cdot i_b} = -\beta \frac{(R_{B2} \parallel R_C)}{r_{be}} = -105$$

$$r_i = R_{B1} \parallel r_{be} = 1k\Omega$$

$$r_o = R_{B2} \parallel R_C = 2.2k\Omega$$

11.



不能

